

The TPC-18 Full-Block Physical Remainder: Exact Range Tests, Canonical Residuals, and Literal Dual Stop Certificates

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Abstract

TPC-18 derives a powerful determinant-dispersion estimate on admissible hyperbola blocks, but its global fixed-shift conclusion is conditional on those blocks covering the full physical tail, or on the uncovered physical remainder being $o(X)$. We isolate that condition as a finite exact range problem.

Let w_X be the literal physical coefficient vector on the complete active atom set, and let

$$B_X : \mathbb{C}^{\mathcal{B}_X} \longrightarrow \mathbb{C}^{\Omega_X}$$

be the synthesis whose columns are the actual TPC-18 blocks with their inherited overlap weights, signs, cutoffs and normalization. Writing $P_X = B_X B_X^\dagger$, we prove

$$w_X \in \text{ran } B_X \iff (I - P_X)w_X = 0 \iff \langle y, w_X \rangle = 0 \quad \text{for all } y \in \ker B_X^*.$$

The canonical coefficient and residual are

$$c_X^* = B_X^\dagger w_X, \quad r_X^* = (I - P_X)w_X,$$

and the physical scalar splits exactly into a block term plus $\nu_X \langle a_X, r_X^* \rangle$. The sharp distance to the block carrier is $\|r_X^*\|$; when it is nonzero, a maximizing unit dual witness lies in $\ker B_X^*$. Thus one nonzero exact witness disproves exact cover, while a nonzero residual is still compatible with the route only if its literal physical pairing is separately $o(X)$.

Exact cover alone is not an arithmetic estimate: transporting a block theorem costs the actual canonical coefficients or the relevant singular-value bound, and that cost enters the physical ledger. The present repository does not contain a machine-readable growing TPC-18 block matrix and coefficient vector from which the range test could be executed. Accordingly the full-block hypothesis remains open. The result is an L0/L1 cover audit, not an L2 fixed-shift saving or a twin-prime theorem.

Keywords: full-block reassembly; exact cover; physical residual; dual witness; least squares; endpoint ledger.

Literal-carrier convention. The row set is required to be the complete physical atom set specified by a verified TPC-116 archive. A column is an actual admissible TPC-18 block, not an abstract block of similar dimensions. Overlap weights, both polarizations, cutoffs, prime-power errors, the fixed $h_0 \neq 0$, and the global normalization are part of the matrix data.

1 The inherited gap

TPC-18 constructs a Type-II tail interface and proves conditional block estimates under a determinant-dispersion hypothesis [3]. Its conditional fixed-shift conclusion assumes an additional global all-block identity

$$H_{h_0, R; V}^{\text{HB}}(X) = \sum_{b \in \mathcal{B}_X} (C_b + T_b) + o(X),$$

including every geometric and normalization requirement. The paper does not prove that global cover. TPC-92 later stresses that an exact full-block cover is sufficient, while any nonzero residual requires a separate physical theorem [4]. TPC-116 returns the same issue after completing the complement taxonomy [2].

The word “cover” can mean three different things:

1. every atom belongs to at least one set-theoretic block;
2. the incidence columns span the coefficient support;
3. the *weighted signed physical vector* belongs to the range of the literal block synthesis.

Only the third statement supplies the identity needed by TPC-18. The first two can fail after overlap weights, masks and signs are restored.

2 The literal finite model

Let Ω be the finite active atom set and equip $\mathcal{H} = \mathbb{C}^\Omega$ with its declared physical inner product. Let $w \in \mathcal{H}$ be the literal coefficient vector. For the finite block set $\mathcal{B}$, define

$$B : \mathbb{C}^{\mathcal{B}} \rightarrow \mathcal{H}, \quad Bc = \sum_{b \in \mathcal{B}} c_b v_b,$$

where v_b contains every atom weight contributed by block b . The matrix is allowed to have repeated, dependent or zero columns. Every adjoint, orthogonal projection, norm and Moore–Penrose inverse below is computed relative to this declared physical inner product and the declared coefficient-space inner product; a hidden change to an unweighted Euclidean geometry changes the certificate. Put

$$P = BB^\dagger, \quad c^* = B^\dagger w, \quad r^* = (I - P)w.$$

Theorem 2.1 (Exact cover alternatives). *The following are equivalent:*

- (i) $w \in \text{ran } B$;
- (ii) $Pw = w$;
- (iii) $r^* = 0$;
- (iv) $\langle y, w \rangle = 0$ for every $y \in \ker B^*$.

When they hold, c^* is the unique least-norm exact cover:

$$Bc^* = w, \quad \|c^*\|_2 = \min_{Bc=w} \|c\|_2.$$

Proof. The Moore–Penrose identity makes $P = BB^\dagger$ the orthogonal projection onto $\text{ran } B$. Hence (i)–(iii) are equivalent. Finite-dimensional orthogonality gives $(\text{ran } B)^\perp = \ker B^*$, proving (iv). The least-norm property of $B^\dagger w$ follows by decomposing any solution into $(\ker B)^\perp \oplus \ker B$. $\square$

Proposition 2.2 (Sharp residual and dual certificate). *For every w ,*

$$\text{dist}(w, \text{ran } B) = \|r^*\|_2 = \max_{\substack{y \in \ker B^* \\ \|y\|_2 \leq 1}} |\langle y, w \rangle|.$$

If $r^ \neq 0$, the maximum is attained by $y = r^*/\|r^*\|_2$. Thus any exact vector $y \in \ker B^*$ with $\langle y, w \rangle \neq 0$ is a literal certificate that the proposed block family does not cover w .*

Proof. Orthogonal projection gives the distance identity. For $y \in \ker B^* = (\text{ran } B)^\perp$, $\langle y, w \rangle = \langle y, r^* \rangle$, so Cauchy–Schwarz gives the upper bound and the stated vector attains it when $r^* \neq 0$. When $r^* = 0$, the vector $y = 0$ belongs to the closed unit ball and attains the value zero. $\square$

Remark 2.3 (A range witness is stronger than a missing-set label). A set-theoretically uncovered atom gives a coordinate witness only when no block column has any component in that coordinate. In general, signed overlap can create subtler cokernel vectors. Proposition 2.2 detects both mechanisms without changing the physical weights.

3 The physical scalar and the allowed remainder

Let $a \in \mathcal{H}$ be the physical evaluation vector and let ν_X be the inherited scalar normalization. The target is

$$H_X = \nu_X \langle a, w \rangle.$$

Theorem 3.1 (Exact block–remainder identity). *For the canonical pair c^*, r^* ,*

$$H_X = \nu_X \langle B^* a, c^* \rangle + \nu_X \langle a, r^* \rangle.$$

Consequently the TPC-18 cover gate closes if either:

- (a) $r^* = 0$ and the complete block term is $o(X)$; or
- (b) the complete block term is $o(X)$ and the literal physical residual satisfies $\nu_X \langle a, r^* \rangle = o(X)$.

No smallness of $\|r^\|_2$ is necessary if the physical pairing is proved small; conversely, $r^* \neq 0$ alone does not prove that its physical pairing is large.*

Proof. Since $w = Bc^* + r^*$, take its inner product with a and use adjointness. The two sufficient alternatives are immediate. $\square$

Corollary 3.2 (A norm-based sufficient residual estimate). *If*

$$\nu_X \|a\|_2 \|r^*\|_2 = o(X),$$

then the physical residual in [theorem 3.1](#) is $o(X)$. This criterion is sufficient and sharp given only the two norms.

Proof. Cauchy–Schwarz proves the bound; collinear vectors show sharpness. $\square$

4 Exact cover is not an estimate

Suppose a block theorem bounds the vector of block evaluations

$$q = B^* a, \quad |q_b| \leq U_b.$$

Even under exact cover,

$$|H_X| \leq \nu_X \sum_{b \in \mathcal{B}} |c_b^*| U_b.$$

Thus the number, size and conditioning of the canonical coefficients matter.

Proposition 4.1 (Coefficient transport costs). *If $r^* = 0$, then*

$$|H_X| \leq \nu_X \|c^*\|_2 \|B^* a\|_2 \leq \nu_X \|B^\dagger\| \|w\|_2 \|B^* a\|_2.$$

Both inequalities have optimal constant one. If instead individual block bounds are inserted, the exact triangle cost is $\sum_b |c_b^| U_b$. Neither cost may be omitted merely because the cover is exact.*

Proof. Use Cauchy–Schwarz, then $\|c^*\| = \|B^\dagger w\| \leq \|B^\dagger\| \|w\|$. Equality examples are supplied by singular vectors and collinear evaluation vectors. $\square$

Remark 4.2 (Relation to TPC-113). TPC-113 identifies the quotient condition number of a literal physical synthesis [1]. Here B is a different, specific full-block synthesis. A previously paid frame cost may be reused only through an exact factorization identifying the same operator event; otherwise $\|B^\dagger\|$ is a new ledger input.

5 What an executable TPC-18 archive must contain

An actual range decision requires:

1. a stable row identifier for every physical atom returned by the complete TPC-116 partition;
2. a stable column identifier for every C_b and T_b , with the allowed L, K, I and primitive/nonprimitive status;
3. exact or certified interval entries including overlap weights, cutoffs, both polarizations, Möbius/von Mangoldt factors and the prime-power correction;
4. the literal vector w_X , evaluation a_X , global normalization ν_X , and fixed h_0 ;
5. a certificate $Bc = w$, or a dual witness $B^*y = 0$, $\langle y, w \rangle \neq 0$, or the canonical residual plus a quantitative physical estimate; and
6. the coefficient/conditioning cost used to transport the block dispersion theorem.

Dimensions and set-theoretic coverage statistics are useful diagnostics but do not replace these fields.

6 Finite exact regression certificate

The script `experiments/tpc117_cover_certificate.py` uses `fractions.Fraction` to verify on an overlapping block matrix:

1. an exact cover $Bc = w$;
2. an exact cokernel vector $B^*y = 0$;
3. a perturbed vector with nonzero dual pairing;
4. the canonical rational residual and distance identity, including the full-row-rank case $\ker B^* = \{0\}$; and
5. the exact physical block-remainder decomposition.

The example demonstrates the certificate format only. It is not a sample of the growing TPC-18 carrier.

7 Stop/go verdict and claim boundary

Status	Certificate	Consequence
Go exact	Literal archive and verified $B_X c_X = w_X$, with affordable transport cost	No physical cover residual; pass the charged block estimate to the endpoint ledger.
Go with remainder	Canonical identity plus $\nu_X \langle a_X, r_X^* \rangle = o(X)$	The nonzero residual is separately controlled at the original normalization.
Stop the exact-cover subroute	Exact $B_X^* y_X = 0$ and $\langle y_X, w_X \rangle \neq 0$, with no residual theorem	Exact cover is false; enlarge/change the carrier or prove the physical residual small.
Current TPC state	TPC-18 states the global cover as an additional hypothesis; no machine-readable growing matrix/vector certificate is available here	H6 is mathematically OPEN; its present range test is NOT-TESTABLE-FROM-CURRENT-ARTIFACTS, and its ledger cost is unknown.

Established. Theorems 2.1 and 3.1 and propositions 4.1 and 2.2 are exact L0 finite-dimensional theorems. The required literal TPC-18 archive schema and original-scale pairing are L1 interfaces.

Not established. We do not prove that the actual TPC-18 blocks cover the growing physical vector, and we do not bound their canonical residual or transport cost. No determinant-dispersion hypothesis is upgraded. There is no new L2 fixed- h_0 estimate, parity breakthrough, Hardy–Littlewood asymptotic, prime-pair lower bound or twin-prime theorem. The next gate is the nonduplicated strict $1/400$ endpoint ledger.

References

- [1] Liang Wang. Canonical Physical Frame and Reassembly Cost: Exact Quotient Singular Values, a Sharp Coherence Obstruction, and the Strict Endpoint Gate. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-113-canonical-physical-frame-reassembly, 2026.
- [2] Liang Wang. Complete High-Frequency and Ultra-Tail Closure at the Original Physical Normalization. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-116-complete-tail-closure, 2026.
- [3] Liang Wang. Hyperbola-Block Rigidity and Determinant Dispersion: A Type-II Route Beyond the Möbius Tail. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-18-mobius-tail-determinant-dispersion, 2026.
- [4] Liang Wang. A Ten-Paper Mass–Cancellation–Cover Audit: Exact Closures, Independent Arithmetic Gates, and a Stop/Go Roadmap. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-92-ten-paper-mass-cancellation-cover-audit, 2026.